Natural Resources and Environmental Monitoring

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Workshop
on
Combining Satellite and In-Situ Data for Ecosystem Monitoring

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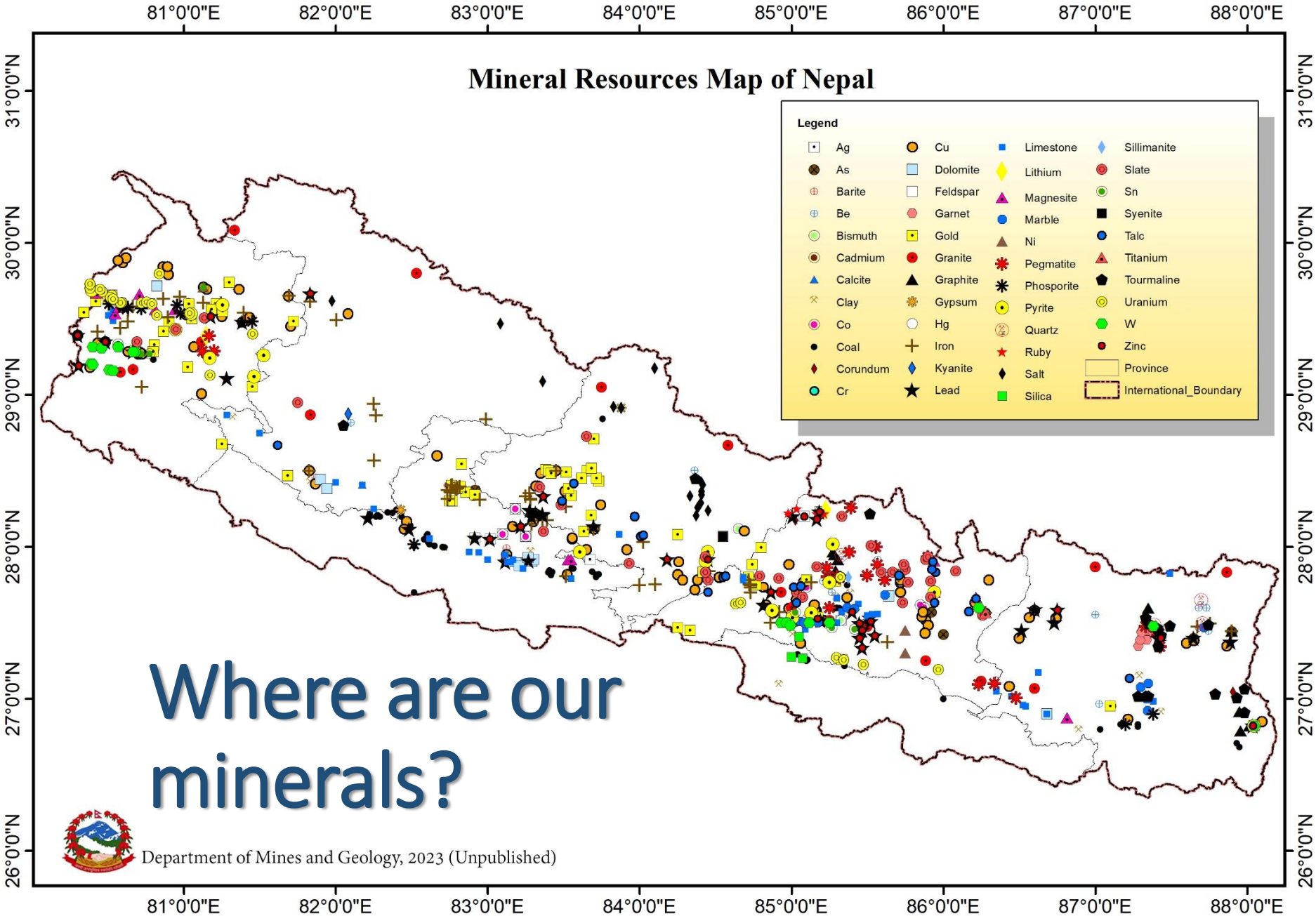
Natural Resources of Nepal

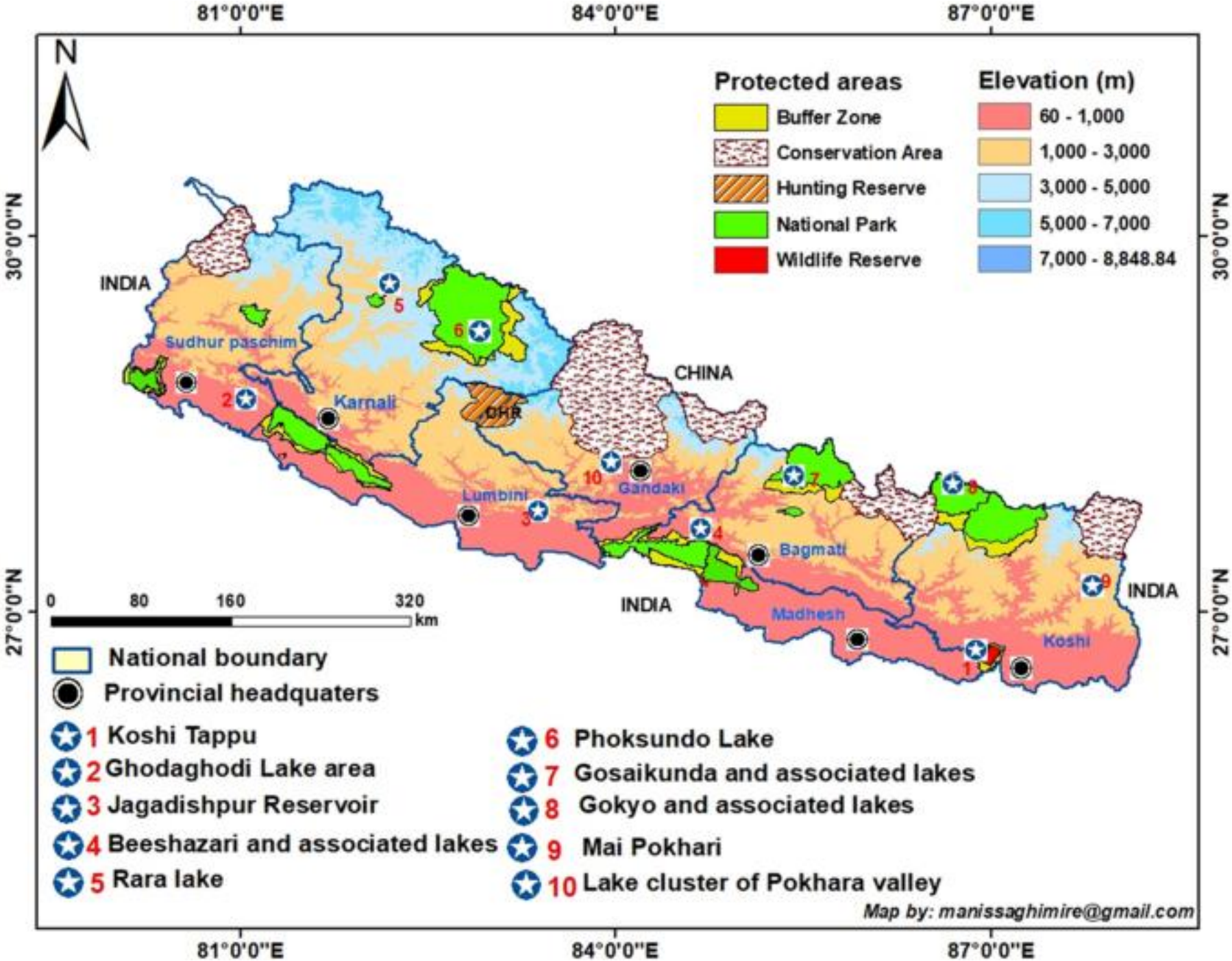
- **Immense Resource Base Country**
  - Mountains, Water, forests, Biodiversity, Agriculture, Landscapes, Heritage…

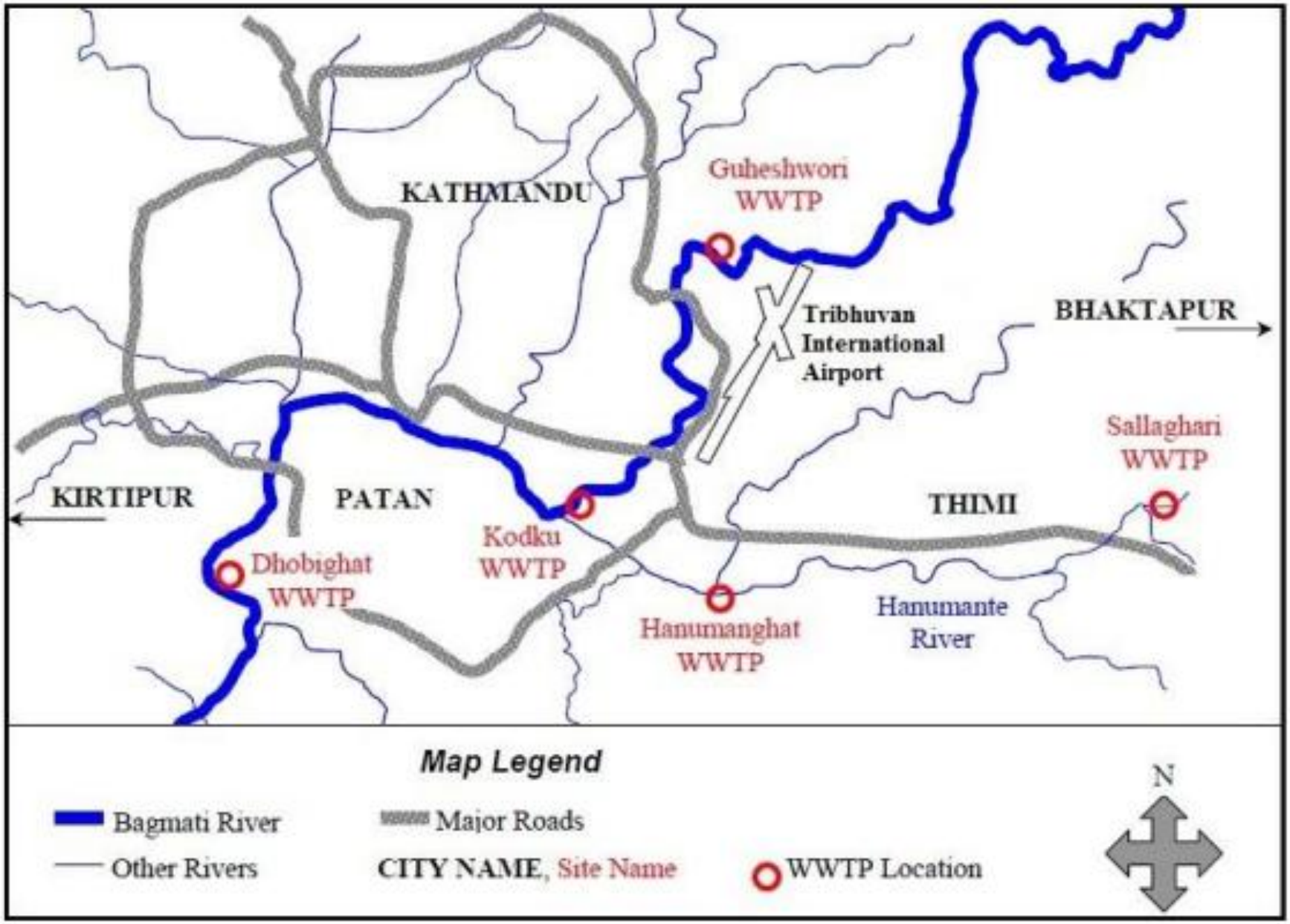
Unexplored Natural Wealth

- **Critical Questions**
  - Where are our resources?
  - What is the quantity and quality of the water reserves?
  - What is the actual rate of forest biomass change or wildlife threat?
  - How are these resources responding to a changing climate?
  - E.g., Is shifting monsoon patterns affecting the water availability?



Where are our minerals?


81°0'0"E
84°0'0"E
87°0'0"E
30°0'0"N
27°0'0"N
N
Protected areas
Buffer Zone
Conservation Area
Hunting Reserve
National Park
Wildlife Reserve
Elevation (m)
60 - 1,000
1,000 - 3,000
3,000 - 5,000
5,000 - 7,000
7,000 - 8,848.84
INDIA
CHINA
Sudhur paschim
Karnali
DHR
Lumbini
Gandaki
Bagmati
Madhesh
Koshi
0 80 160 320 km
National boundary
Provincial headquaters
1 Koshi Tappu
2 Ghodaghodi Lake area
3 Jagadishpur Reservoir
4 Beeshazari and associated lakes
5 Rara lake
6 Phoksundo Lake
7 Gosaikunda and associated lakes
8 Gokyo and associated lakes
9 Mai Pokhari
10 Lake cluster of Pokhara valley
Map by: manissaghimire@gmail.com



https://www.theseus.fi/bitstream/handle/10024/59525/Regmi_Shakil.pdf?sequence=1

Challenge of Monitoring

- **Inaccessible Terrain**
  - Mostly in high-altitude or remote areas and are difficult or expensive to reach and monitor consistently
- **Sparse and Disconnected Data**
  - The monitored data are limited, distributed, disconnected and sometimes inaccessible
- **Complex and Dynamic Systems**
  - Our ecosystems are too complex and vary from small scale to large landscape
- **Finance**
  - Priority of the state (Basic needs!)

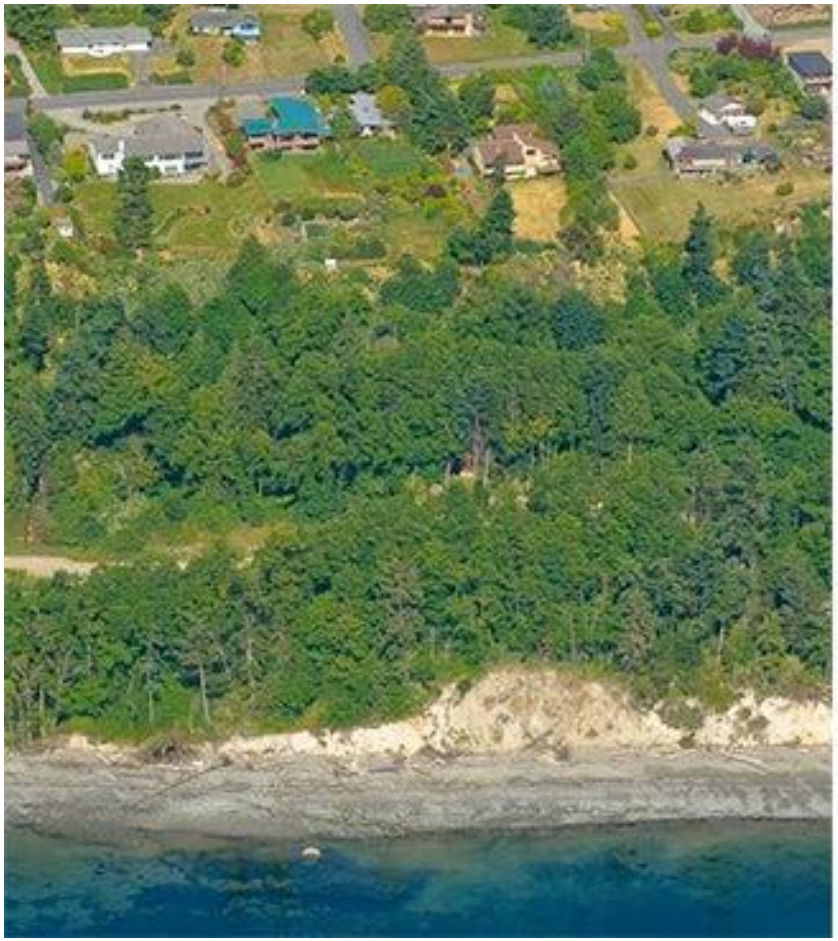
Combining In-Situ Data with Satellite Remote Sensing

- **In-Situ Data**
  - The actual local data
  - Measuring points
  - Monitoring stations
- **Satellite Data**
  - Large coverage
  - Time series
  - Optical, radar, and thermal imagery

The power lies in their combination

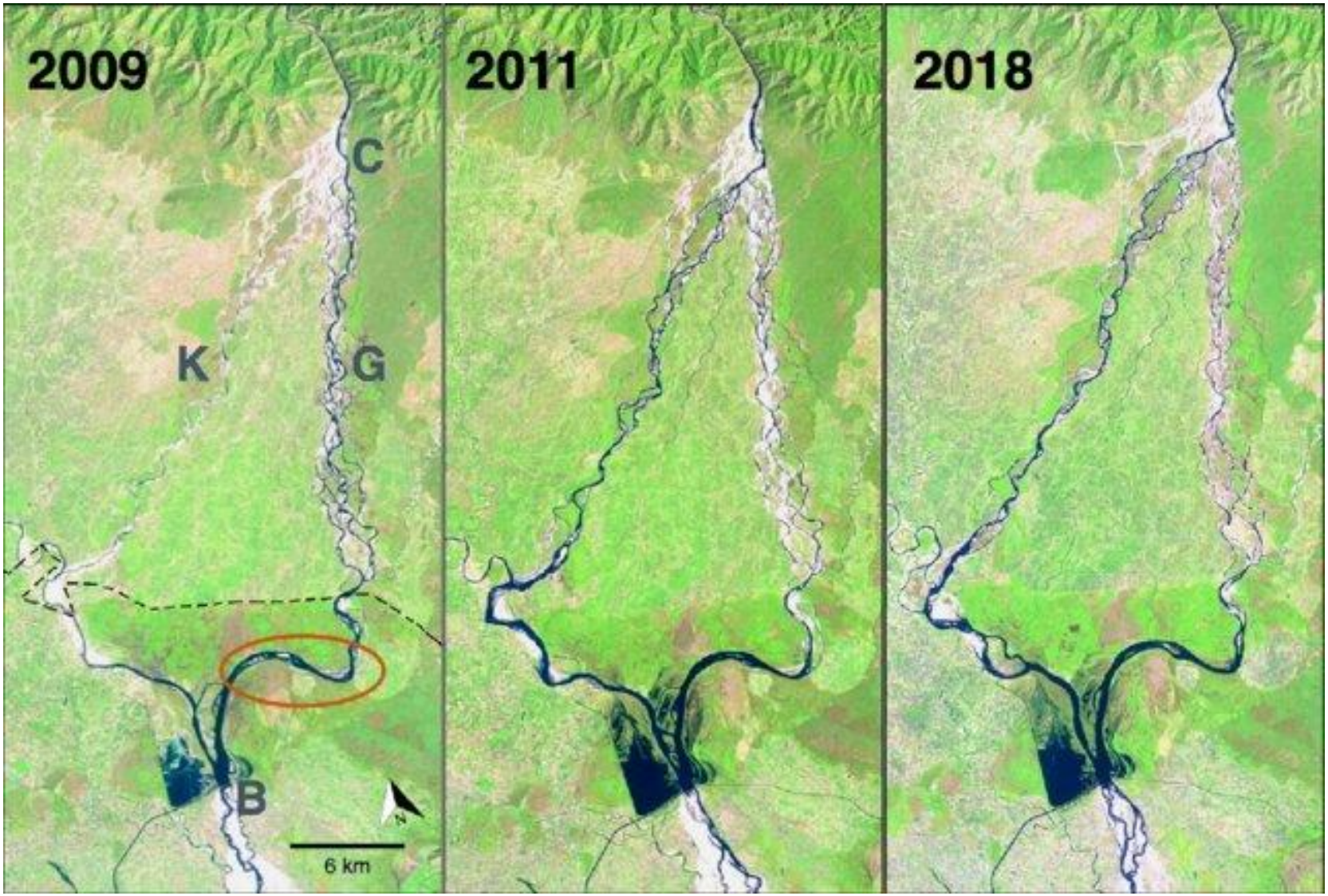
Data to Decisions

- **Resource Exploration**
- **Environmental Management**
- **Disaster Risk Reduction**
- **Climate Adaptation and Sustainability**



A combination photo shows the same area near State Route 530 before and after the mudslide that devastated the community of Oso, Wash.

A photo from Google is seen next to a photo that was taken on March 24, after the massive mudslide that killed at least 14 people Saturday and left dozens missing. Google (left) | Ted S. Warren / AP (right)



https://www.researchgate.net/profile/Gaurav-Vashistha

Sustainability through monitoring

- Let the mysterious resources be known and plan for their best uses
- Let combine the ground data with historical or satellite data to overcome uncertainty, risks or damages and for future prediction
- Let combine the individual effort to a group strength for a sustainable environment